

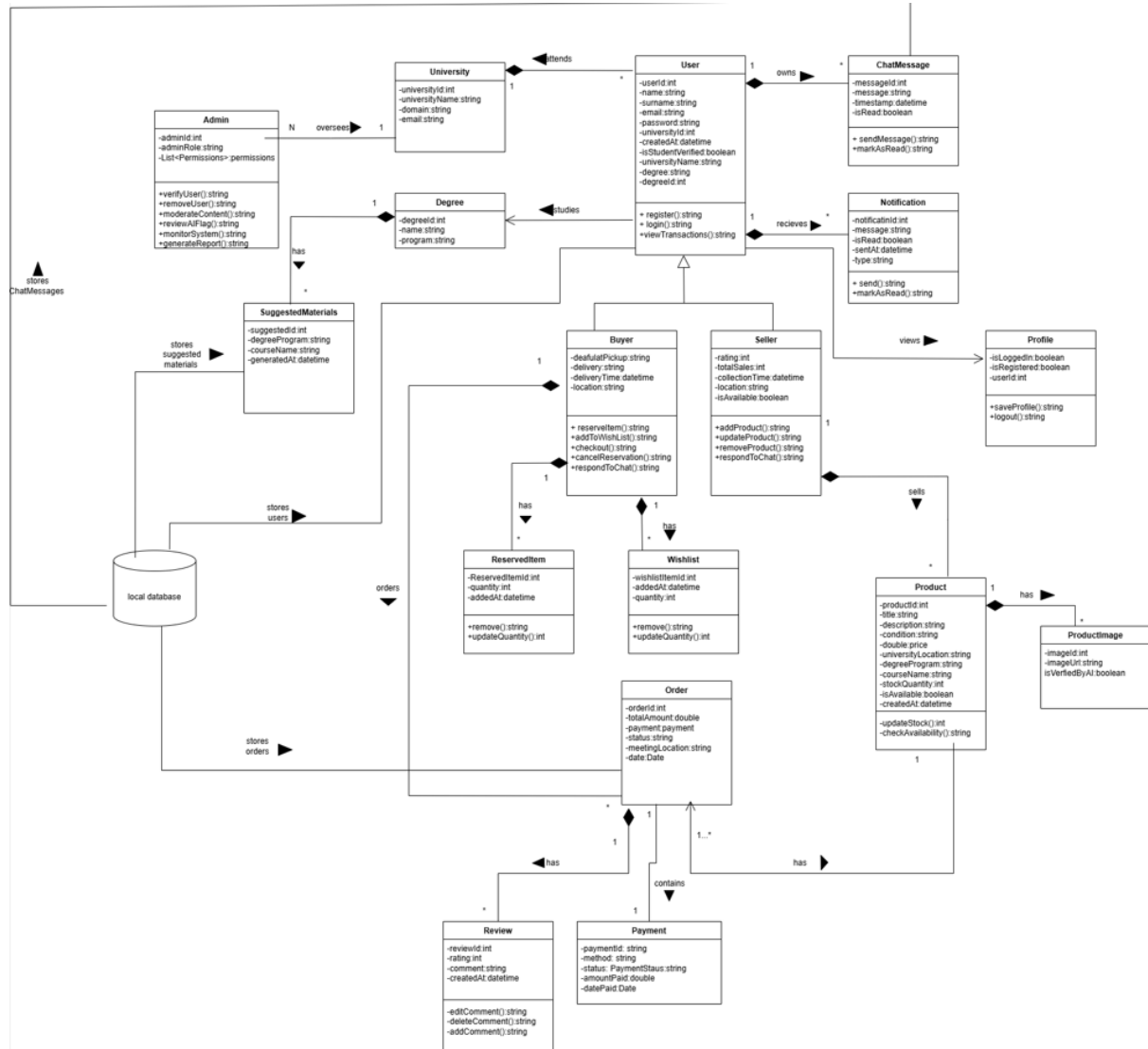
UNITRADE

Introduction

Scope

UniTrade is a web and mobile platform designed to connect university students who want to buy and sell second-hand academic materials. Built for the South African university context, where students face significant financial pressure from high tuition fees and expensive prescribed textbooks, the system provides a safe, verified, and cost-effective marketplace for books and study resources. Currently, students either purchase new materials at full price or rely on informal networks to find second-hand options, which is unreliable and often unsafe. UniTrade solves this by introducing a structured platform where students can list, browse, and purchase used materials directly from verified peers. The system includes AI-assisted listing verification, OTP-based student authentication, real-time chat, GPS meetup coordination, and OZOW payment processing.

Domain Model



Functional Requirements

R1: User and identity management

R1.1: The system shall allow a user to register.

R1.1.1: The system shall allow a registered university student to be registered in the system.

R1.1.2: The system shall provide an input area for student email, student number and proof of registration pdf.

R1.1.3: The system shall verify that the user is a student at said university by sending an otp to the user's university email.

R1.1.4: The system shall allow users to resend otp to their email if it expires.

R1.1.5: The system shall extract text from uploaded proof of registration pdf and compare the student number and degree against the registration form data using AI.

R1.1.6: The system shall assign a confidence score(0-100) to the ai verification result.

R1.1.7: The system shall provide manual verification of students' degrees if AI fails or gives a score below 75.

R1.1.8: The system must be able to provide users with a verification status.

R1.1.9: The system shall allow users to resubmit proof of registration pdf if manual verification fails.

R1.2: The system shall allow users to remain logged into the website or mobile application.

R1.3: The system shall allow Logged in users to reset their password.

R1.4: The system shall allow users to switch roles between being a seller or buyer.

R1.5: The system shall allow logged in users to log out.

R1.6: The system shall restrict unverified users to read-only access.

R1.7: The system shall prevent unverified users from initiating chats

R1.8: The system shall prevent unverified users from making purchases.

R1.9: The system shall store listings from unverified sellers as drafts only.

R2: Listing and inventory management.

R2.1: The system shall allow sellers to create a listing draft

R2.1.1: The system shall allow users to upload images of materials.

R2.1.2: The system shall allow users to provide details of the product they are selling.

R2.1.3: The system shall use AI to assess uploaded images for object recognition (must identify as a book or educational material) before it is published.

R2.1.4: The system shall assign a quality score(0-100) to each image.

R2.1.5: The system shall automatically approve users when AI confidence score exceeds threshold.

R2.1.6: The system shall have the admin intervene to manually check the quality of materials score is below threshold.

R2.1.7: The system shall classify listings into low, medium, and high risk categories.

R2.1.8: The system shall apply different visibility rules based on risk level.

R2.2: The system shall allow sellers to edit their listings.

R2.3: The system shall allow sellers to remove their listings

R2.4: The system shall allow sellers to view all their listings.

R2.5: The system shall allow sellers to save their listing so it's in the process of getting published.

R2.6: The system shall keep all new listings in a draft state until submission.

R2.7: The system shall prevent unverified sellers from submitting listings.

R2.8: The system shall publish listings automatically for low-risk scores.

R2.9: The system shall reduce visibility of medium-risk listings.

R2.10: The system shall prevent high-risk listings from being published.

R2.11: The system shall allow sellers to unreserve the listing the buyer has reserved on their material.

R3: Discovery and recommendations

R3.1: Users that are in buyer mode must be able to view their degree specific dashboard.

R3.1.1: The system shall provide users with suggested materials.

R3.2: The system shall allow buyers to search for materials

R3.2.1: Buyers can search by seller name.

R3.2.2: Buyers can search listings by material title.

R3.2.3: Buyers can search listings by course code.

R3.3: The system shall allow users to filter results.

R3.3.1: Buyers shall be able to filter search results by price range.

R3.3.2: Buyers shall be able to filter search results by date listed.

R3.3.3: The system shall allow buyers to view details of the material

R3.4: The system shall allow buyers to reserve listings.

R3.4.1: Buyers must be able to proceed to checkout.

R3.5: The system shall allow buyers to add material to wishlist

R4: Reservation and Transaction

R4.1: The system shall allow buyers to reserve items.

R4.1.1: The system shall allow buyers to reserve items for a 24 hours window period.

R4.1.2: The system shall require seller acknowledgement within 24 hours.

R4.1.2: The system shall allow buyers to reserve a full degree list of materials.

R4.1.3: The system shall allow buyers to reserve the full degree package for days.

R4.1.4: The system shall allow sellers to manually release reservations after 12 hours of buyer inactivity.

R4.1.5: The system shall pause reservation timers when both users are actively communicating.

R4.1.5: The system shall provide an auto cancellation for reservations that surpass the given time frame of the material.

R4.1.6: The system shall expire reservations if either party is inactive.

R4.2: The system shall provide a payment gateway to process buyers payments.

R4.3: The system shall allow buyers to confirm payment once they've met up with the seller.

R4.3.1: The system shall generate a pin for the buyer.

R4.3.2: The system shall allow sellers to enter a pin given by the buyer during their meet-up.

R4.4: The system shall allow buyers to raise a dispute if materials were not as they were displayed.

R5: Admin and Dispute Resolutions:

R5.1: The system shall allow admins to view pending verifications.

R5.1.1: Admin must be able to approve/reject proof of enrollment.

R5.2: The system shall allow admins to view all failed listings.

R5.2.1: Admins must be able to manually reject/approve materials.

R5.3: The system shall allow admins to view dispute cases,

R5.3.1: Admins must be able to review evidence.

R5.3.2: Admins must be able to suspend malicious users.

R5.3.3: Admins must be able to flag users as a warning.

R5.4: The system shall provide an admin interface to submit and store POPIA compliance documentation.

R6: Communication And Logistics

R6.1: The system shall provide a real-time chat system where sellers can communicate with buyers

R6.1.1: Users must be able to open a chat thread.

R6.1.1.1: Users must be able to send a message.

R6.1.1.2: Users must be able to get a notification from other users.

R6.2: Users must be able to share their live location.

R6.2.1: Users must be able to select a meeting point from the provided map interface(google maps).

R6.2.2: The system shall allow either party to propose a meeting point, the other party must accept before it is confirmed.

R6.3: The system shall allow users to add a review once a transaction has been completed.

Quality Requirements

QR1: Performance

QR1.1: The system needs to return product search results within 2 seconds under normal load (approximately 100 concurrent users).

QR1.2: The system needs to ensure that page loading times and product listings are less than 3 seconds on a 4G network.

QR1.3: Checkout procedures (payment confirmation) need to be completed within 5 seconds in 95% of cases.

QR1.4: Real-time chat messages shall be delivered within 500 milliseconds under normal network conditions.

QR1.5: Push notifications shall be delivered within 3 seconds of the triggering event (e.g. buyer expresses interest in a listing).

QR2: Reliability

QR2.1: The system needs to be available 99% of the time on a monthly basis.

QR2.2: The system needs to recover from server failure within 60 seconds using auto restart or failover mechanisms.

QR2.3: If data loss during a crash occurs then it needs to be limited to less than 1 transaction lost as regular backups and atomic operations will be implemented.

QR2.4: In the event of AI verification service failure, the system shall automatically fall back to manual admin review without disrupting user access to other features.

QR3: Scalability

QR3.1: The system shall support a minimum of 500 concurrent users during peak periods (e.g. semester start) without degradation in response time.

QR3.2: The system shall be architected to scale horizontally on Azure App Services to accommodate growth in user base without requiring significant re-architecture.

QR3.3: The AI verification service shall be deployed as an independent service to allow it to scale separately from the core application during periods of high listing activity.

QR4: Security

QR4.1: All users must be authenticated via a university email OTP mechanism before gaining any access to the platform.

QR4.2: All sensitive data including personal information and payment details shall be encrypted.

QR4.3: The system shall comply fully with the Protection of Personal Information Act (POPIA) of South Africa, including data minimisation, user consent, and the right to access and delete personal data.

QR4.4: All payment transactions shall be processed through OZOW. The UniTrade application shall not store raw card or banking details at any point.

QR4.5: The system shall implement role-based access control (RBAC) to ensure buyers, sellers, and administrators can only access functionality appropriate to their role.

QR5: Usability

QR5.1: A new user shall be able to complete registration, browse listings, and add an item to cart within 5 minutes without requiring any training or documentation.

QR5.2: The application shall conform to WCAG 2.1 Level AA accessibility standards, including:

- Minimum colour contrast ratio of 4.5:1 for normal text
- All images shall include descriptive alt text
- All form fields shall be clearly labelled
- Error messages shall be descriptive and actionable

QR5.3: The web interface shall be fully responsive across desktop, tablet, and mobile screen sizes.

QR5.4: The mobile application shall support both iOS and Android platforms.

QR5.5: The GPS meetup feature shall provide a text-based address alternative for users who have difficulty interacting with map interfaces.

QR6: Maintainability

QR6.1: All source code shall follow agreed team coding standards and be reviewed via pull requests before merging into main branches.

QR6.2: Automated CI/CD pipelines shall be configured via GitHub Actions or Azure DevOps to deploy to development and User Acceptance Testing environments on every approved merge.

QR6.3: The system shall implement centralised logging and error monitoring using Azure Monitor to enable rapid fault diagnosis.

QR7: Administration Requirements

QR7.1: An administrator shall complete the manual verification of a student's proof of registration within 2 business days of the verification request being submitted.

QR7.2: The system shall send the student an automated notification if their verification has not been processed within 1 business day, informing them of the expected resolution time.

QR7.3: An administrator shall resolve a buyer-seller dispute within 5 business days of the dispute being raised.

QR7.4: The system shall automatically notify both parties involved in a dispute of the admin's final decision within 24 hours of the decision being made.

QR7.5: The system shall automatically freeze any funds related to a dispute immediately upon the dispute being raised, pending admin resolution.

QR7.6: If a buyer does not confirm receipt or raise a dispute within 48 hours of the agreed meetup time, the system shall automatically release funds to the seller and close the transaction.

QR7.7: The admin dashboard shall display all pending verifications and active disputes sorted by submission date to ensure oldest cases are prioritised first.

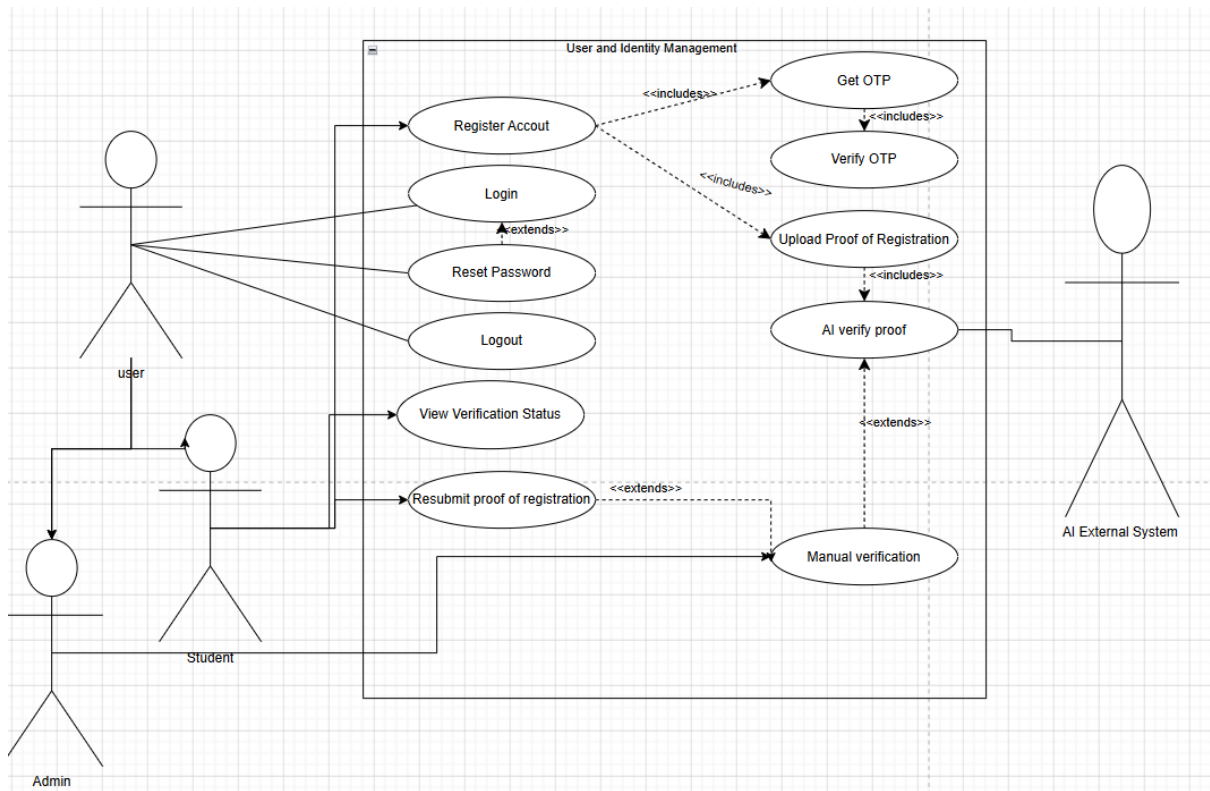
QR8: Legal and Regulatory Compliance

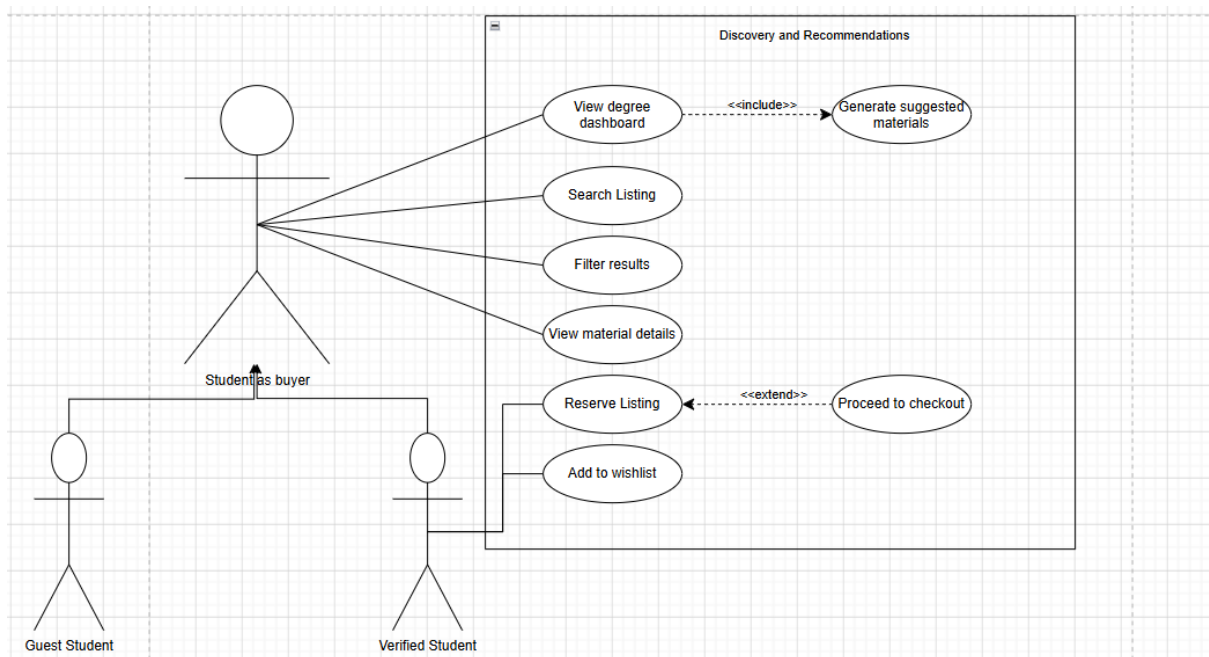
QR8.1: The system shall comply with POPIA for all personal data handling.

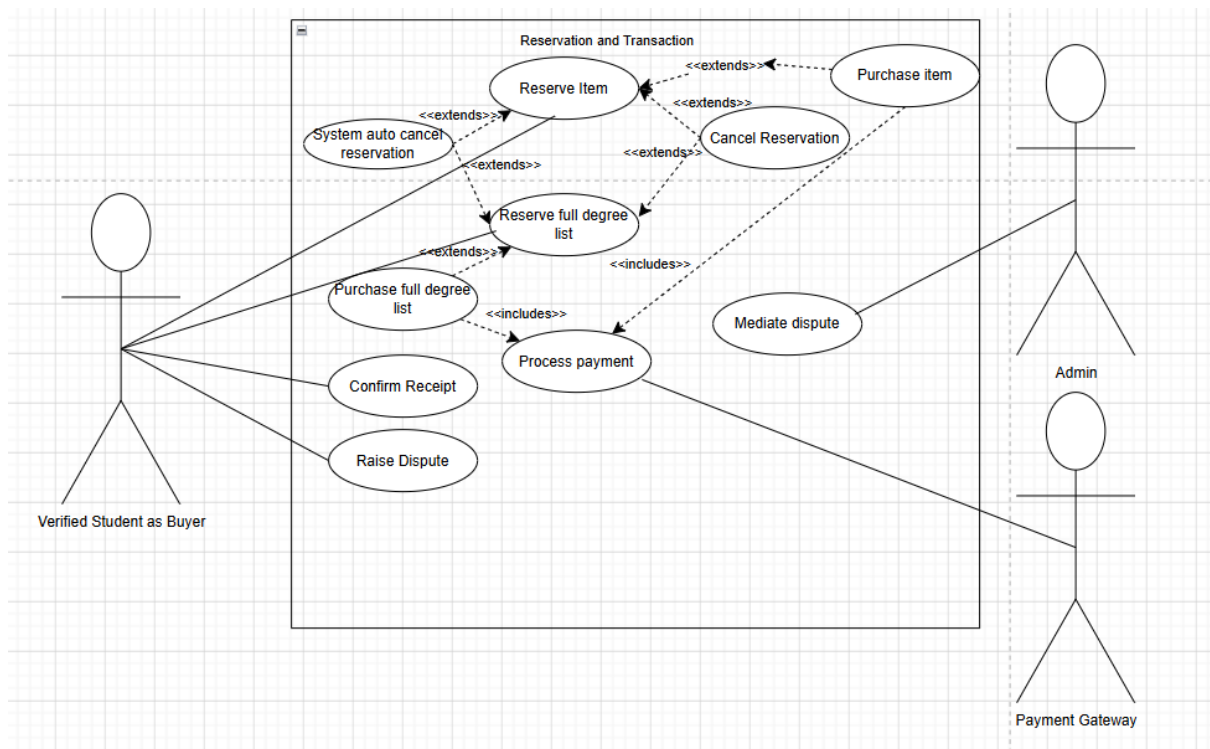
QR8.2: The system shall include clear Terms of Service and Privacy Policy accessible to all users before registration.

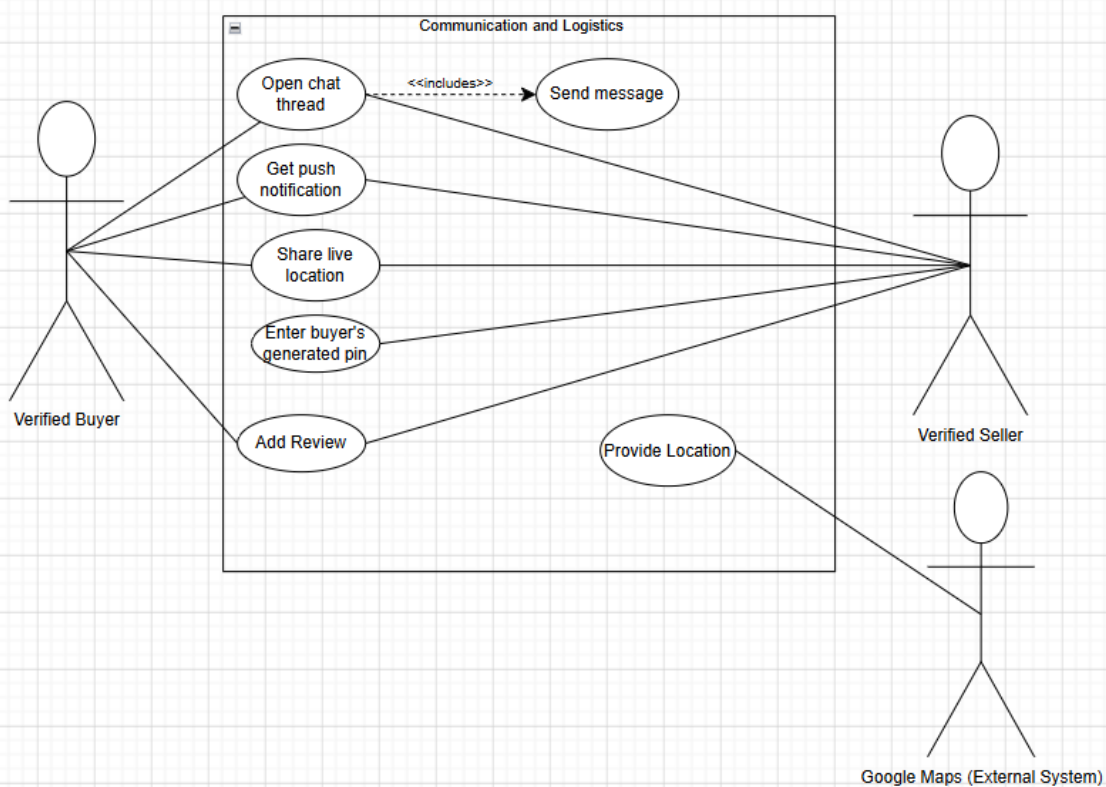
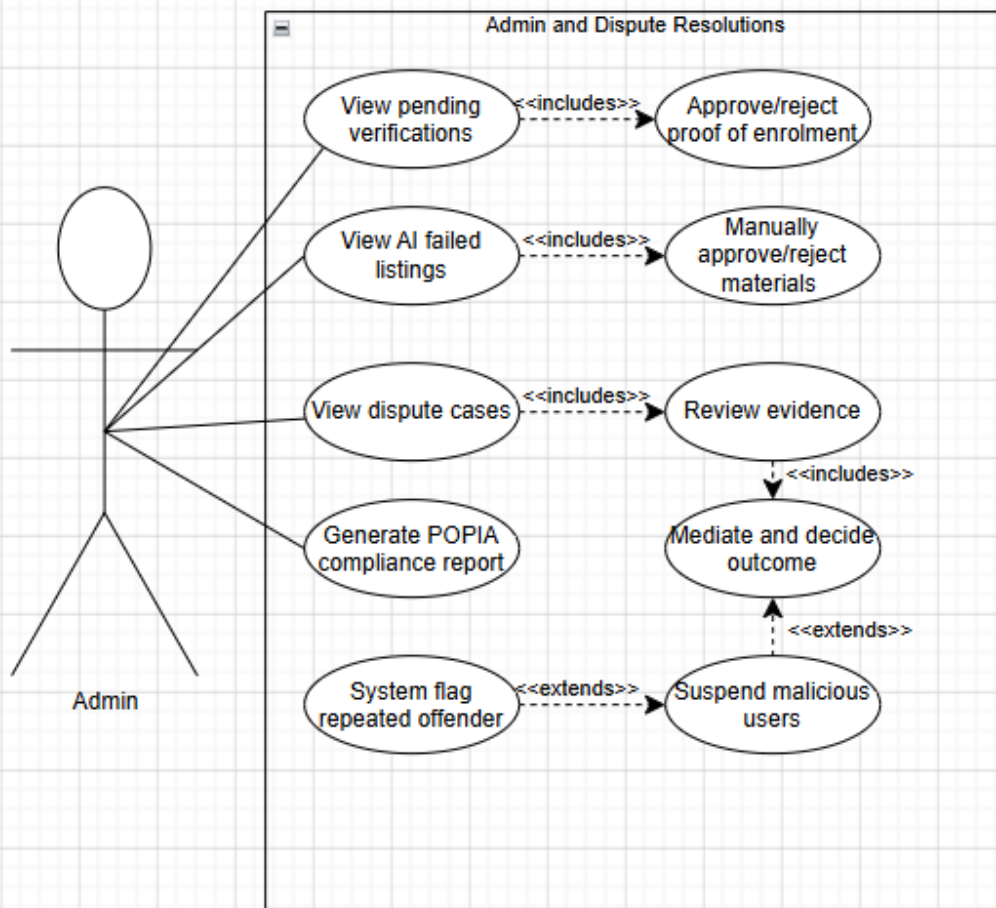
QR8.3: The system shall maintain an audit trail of all transactions, disputes, and admin decisions for a minimum period of 3 years in compliance with South African financial record-keeping requirements.

Use Case Diagrams









Epic, user stories and acceptance criteria

Epic 1: User and Identity Management

Enable South African university students to register, authenticate, and verify their student status so that the marketplace remains exclusive to legitimate students.

US-1.1 — Register an account

As a prospective student user I want to register with my name, surname, university, degree, year of study, student email, and password so that I can begin the process of accessing the marketplace.

Acceptance Criteria:

- Given I am on the registration screen, when I submit valid details with a recognised SA university email domain, then my account is created in pending verification state.
- Given I submit an email with a non-SA-university domain, when I attempt to register, then registration is rejected with a clear message naming the domain check as the reason.
- Given I submit an email already registered, when I attempt to register, then I see a message prompting me to log in or reset my password (the system does not reveal whether the account is verified).
- Given I am on the registration screen, when I submit the form with missing or invalid required fields, then I see field-level error messages and the form is not submitted.

US-1.2 — Receive and verify OTP

As a newly registering user I want to receive a one-time PIN on my student email and enter it so that the system confirms I control the email address.

Acceptance Criteria:

- Given my account was just created, when registration completes, then an OTP is sent to my student email within 60 seconds.
- Given I enter the correct OTP within its validity window, when I submit, then my email is marked verified and I proceed to the proof-of-registration step.
- Given I enter an incorrect OTP, when I submit, then I see an error and remain on the OTP screen; after a configured number of failed attempts, the OTP is invalidated and I must request a new one.

- Given the OTP has expired, when I submit it, then the system rejects it and offers to resend.
- Given I request a resend, when triggered, then a new OTP is sent and the previous one is invalidated; resends are rate-limited.

US-1.3 — Upload proof of registration

As a verified-email user I want to upload my proof of registration as a PDF so that the system can verify I am a current student.

Acceptance Criteria:

- Given I have verified my email, when I reach the upload step, then I can select a PDF file from my device.
- Given I attempt to upload a non-PDF file, when I submit, then the upload is rejected with a message stating PDF is required.
- Given the file exceeds the maximum size, when I submit, then the upload is rejected with a clear size limit message.
- Given a successful upload, when complete, then my proof is queued for AI verification and my status updates to under review.

US-1.4 — AI-assisted proof verification (system-driven)

As the system I want to automatically score uploaded proofs of registration for legitimacy so that confident-positive cases are auto-approved without admin workload.

Acceptance Criteria:

- Given a PoR has been uploaded, when AI scoring runs, then it returns a confidence score and a routing decision: auto approve, manual review, or manual review high risk.
- Given the score is in the auto-approve band, when scoring completes, then the user's account is upgraded to verified and the user is notified in-app and by email.
- Given the score is in the manual-review band, when scoring completes, then the case is added to the admin verification queue and the user remains in under review.
- Given the score is in the manual-review-high-risk band (clear forgery indicators), when scoring completes, then the case is also routed to the admin queue with a high risk flag — the AI never auto-rejects; rejection is always a human decision.
- Given a user has uploaded a PoR, when AI scoring is in progress, then the user sees a loading indicator until a result is returned.

- Given a PoR has been scored, when the result is stored, then the system records the score, decision, timestamp, and model version for audit purposes.

US-1.5 — Manual verification by admin

As an admin I want to review proofs of registration that AI flagged for manual review so that legitimate students get verified and forgeries are rejected.

Acceptance Criteria:

- Given a case is in the admin verification queue, when I open it, then I see the uploaded PoR, the user's submitted details, the AI score, and any AI-flagged reasons.
- Given I approve the case, when I confirm, then the user's account is upgraded to verified and they receive an in-app and an email notification.
- Given I reject the case, when I confirm, then I must select a terminal rejection reason (forged, expired, wrong document type, other) and add a free-text note; the user is notified and cannot resubmit without admin intervention.
- Given I request changes, when I confirm, then I must select a request changes reason (unreadable, missing information, other) and add a free text note; the user is notified with the reason and prompted to resubmit.
- Given I perform any verification action (approve, reject, request changes), when the action is confirmed, then the system records the admin ID, action taken, timestamp, and reason for audit purposes.
- Given a case has exceeded the defined SLA threshold, when I view the verification queue, then the case is prioritised and surfaced at the top of the queue.

US-1.6 — Resubmit proof of registration

As a user whose PoR was rejected or required changes I want to upload a new PoR so that I can complete verification.

Acceptance Criteria:

- Given my PoR is in the changes requested state, when I log in, then I see a clear prompt to resubmit and the reason for the previous rejection.
- Given my PoR is in the changes requested state, when I check my emails, then I see a clear reason why my submission was declined and an instruction to resubmit.
- Given my PoR has been rejected after a manual admin review due to legitimacy concerns (e.g., forgery), when I attempt to access the application, then I am

denied access and I receive both an in-app rejection notice and an email explaining the reason for the decision.

- Given my PoR is in the changes requested state and the request was made by Admin A, when I resubmit a new PoR, then the resubmission is assigned to Admin A for review (not a different admin).

US-1.7 — View verification status

As a registered user I want to see my current verification state at any time so that I know what I can and can't do on the platform.

Acceptance Criteria:

- Given I am logged in, when I open my account/profile area, then I can see my current state: pending email verification, pending por upload, under review, verified, rejected, or changes requested.
- Given I am in under review, when I view my status, then I see an estimated time remaining (based on SLA) and a note that drafts can be created but listings won't go live until verified, and I can browse listings.
- Given I am in rejected or changes requested, when I view my status, then I see the reason and any next-step actions.

US-1.8 — Login

As a registered student I want to log in with my student email and password so that I can access my account.

Acceptance Criteria:

- Given I enter correct credentials, when I submit, then I am authenticated and routed to the appropriate landing page based on my verification status.
- Given I enter incorrect credentials, when I submit, then I see a generic "invalid email or password" message.
- Given I am logged in and idle (no interaction) for longer than the configured inactivity timeout (e.g. 30 minutes), when I attempt to access a protected page, then my session is expired and I am redirected to the login page.
- Given I check the "remember me" box during login, when I close and reopen my browser or return after a long period (but within the extended session lifetime), then I am still required to re-authenticate

US-1.9 — Reset password

As a student who has forgotten my password I want to reset it via an emailverified flow so that I can regain access.

Acceptance Criteria:

- Given I select “forgot password,” when I enter my registered email, then a password reset OTP is sent to my email.
- Given I submit a valid OTP within its validity window, when I enter a new password, then my password is updated and I am logged in.
- Given the OTP has expired or been used, when I attempt to use it, then it is rejected and I am prompted to start the reset flow again.

US-1.10 — Logout

As a logged-in student I want to log out so that my session ends and my account remains secure on shared devices.

Acceptance Criteria:

- Given I am logged in, when I select logout, then I am successfully logged out of the system.
- Given I have logged out, when I attempt to access any authenticated route, then I am redirected to the login screen.
- Given I have logged out, when I try to perform any authenticated action, then it is rejected as unauthorised.

Epic 2: Listing and Inventory Management

Verified students can create, manage, and remove listings. Every submitted listing is auto-scored by AI; high-risk listings route to admin review before going live. Unverified students can save drafts during the verification window — drafts are invisible to buyers and auto-submit once verification completes. Sellers see every listing’s state at all times.

US-2.1 — Create a Listing Draft

As a logged-in student I want to save my work-in-progress as a draft so that I don’t lose it while composing or while I’m waiting for verification.

Acceptance Criteria:

Given I am a logged-in student (verified or not) creating or editing a listing, when I choose to save as draft even with required fields missing, then the system saves the draft without showing validation errors for required fields. Given I am an unverified

student, when I save a listing as a draft, then the draft is flagged with status "pending verification". Given I am an unverified student with one or more completed drafts that contain all required fields and are flagged for processing but not yet submitted, when my account becomes verified, then each flagged draft is automatically submitted for AI scoring without any manual action from me. Given I am the owner of a saved draft, when I open it from my drafts list, then I can edit any field and save it again as a draft. Given I am the owner of a saved draft, when I choose to delete it, then the draft is permanently removed and no longer appears in my drafts list.

US-2.2 — Submit Listing for Review

As a verified student I want to submit a draft listing into the review pipeline so that buyers can discover and buy it.

Acceptance Criteria:

Given I am a verified student, when I submit a draft and required fields are missing, then the system shows validation errors and does not enter AI scoring. Given I am a verified student and I submit a complete draft, when submission completes, then the listing enters AI scoring and I am notified of the resulting state: live, live with reduced visibility, or under review. Given I open an existing draft and submit it, when submission is successful, then the original draft is consumed by the submission.

US-2.3 — AI Score Listing (system)

As the system I want to score every submitted listing for risk so that low-risk listings go live, medium-risk listings go live with reduced visibility, and high-risk listings route to admin review.

Acceptance Criteria:

Given a submitted listing receives a low-risk score, when scoring completes, then the listing is published with full visibility. Given a submitted listing receives a medium-risk score, when scoring completes, then the listing is published with reduced visibility (e.g. lower search rank, hidden from recommendations) and queued for non-urgent admin review. Given a submitted listing receives a highrisk score, when scoring completes, then the listing is not published, is held in the admin review queue, and that queue is prioritised over medium-risk background reviews. Given a listing has been scored, when scoring finishes, then the system logs: timestamp, risk score, flagged reasons, and the model version used.

US-2.4 — Manual Review

As an admin I want to review AI-flagged listings so that legitimate ones reach buyers and problematic ones don't.

Acceptance Criteria:

Given a listing is in the admin review queue, when I open it, then I see the listing details, photos, AI score, AI-flagged reasons, and the seller's history. Given I approve a flagged listing, when I confirm, then the listing is published and the seller is notified. Given I reject a flagged listing, when I confirm, then I must select a reason and add a free-text note; the listing is removed from the platform and the seller is notified with the reason. Given I request changes on a flagged listing, when I confirm, then I must provide a reason; the listing is returned to the seller as a draft (not live) accompanied by the reason, and the seller can edit and resubmit. Given I make any decision (approve, reject, request changes), when I confirm, then the system logs: admin ID, timestamp, outcome, and the reason text provided.

US-2.5 — Edit Active Listing

As a seller I want to edit one of my live listings so that I can correct or update it.

Acceptance Criteria:

Given my listing is reserved (a buyer has an active reservation), when I attempt to edit it, then I can only modify non-material fields (typo fixes, description clarifications); price, photos, and condition are locked. Given I edit a live listing (whether reserved or not), when I save the edit, then the listing is re-scored by AI and may result in reduced visibility or admin review; existing reservations remain valid and are not cancelled. Given I save any edit, when the edit is applied, then the system logs the change (timestamp, old value, new value, my user ID).

US-2.6 — Remove Listing

As a seller I want to remove a listing from circulation so that it no longer appears on the marketplace.

Acceptance Criteria:

Given I am removing a listing, when the removal flow begins, then I am presented with a reason picker (sold off-platform, no longer for sale, other). Given my listing is currently reserved by a buyer, when I attempt to remove it, then I must confirm twice (checkbox + confirm button), and the buyer is automatically notified of the removal.

US-2.8 — View All Listings

As a seller I want to see all my listings in one place so that I can manage my inventory.

Acceptance Criteria:

Given I am viewing my listings, when I apply a state filter (draft, live, reduced visibility, under review, removed), then only listings in the selected state are shown. Given I am viewing my listings, then each listing displays its state, AI score (if available), admin

notes (if any), edit history, and reservation history. Given a listing is in a non-obvious state (e.g. reduced visibility, under review), when I view it, then I see an explanation of the state and any next steps. Given I am viewing my listings, when I apply a sort, then listings are ordered by the chosen criterion (date, state, price, expressions of interest).

Epic 3: Discovery and Recommendations

Help verified students find the materials they need — through casual browsing, intelligent search, behaviour-driven recommendations, and a degree-aware planner that anticipates what they'll need across their studies. Surface seller reputation alongside listings so buyers can make informed decisions before expressing interest.

US-3.1 — Browse listings

As a student I want to browse the marketplace without a specific query so that I can casually explore what's available.

Acceptance Criteria:

- Given I access the marketplace without a search query, then I see a paginated or scrollable list of active listings.
- Given I am viewing browse results, then listings are sorted by relevance to my profile.
- Given I am viewing browse results, then each listing displays title, price, condition, primary photo, seller name, and seller rating summary.
- Given I am viewing browse results, when I apply filters (category, price range, condition, course/module, university) or sorting (newest, price asc/desc, relevance), then results update accordingly.
- Given I am an unverified student, when I browse listings, then I can view listings and I see a banner indicating limited access